

Aaron Legg

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Education

Cornell University, College of Engineering <i>B.S. Chemical Engineering and Operations Research</i>	Ithaca, NY <i>Aug 2024 – May 2028</i>
Scottsdale Community College <i>A.S. General Engineering, Mathematics, General Studies</i>	Scottsdale, AZ <i>Aug 2023 – May 2024</i>
Relevant Coursework: Probability & Statistics I & II, Data Structures, Stochastic Calculus & Processes, Foundations of AI Decisions	

Projects

HMM Regime Trading Bot ↗	Jan 2025
- Built a modular, vectorized Python backtesting engine with Gaussian-HMM forward-algorithm filtered inference and BIC model selection, eliminating look-ahead bias across 10 equities - Capped max drawdown under 10% via a volatility-rank allocation algorithm (60–95% exposure, 1.25x leverage) across 10 equities - Shipped a separate risk layer with tiered circuit breakers, 60-day correlation caps, and 100-run Monte Carlo cash-gap stress testing	
Monte Carlo Option Pricer ↗	Jan 2025
- Built a high-performance European option pricing engine in Java on geometric Brownian motion, hitting 0.01% error versus the Black-Scholes closed form over 1M+ stochastic paths - Improved computation speed 35% by optimizing Box-Muller sampling to O(N) time and O(1) space; validated via put-call parity and EFX futures comparison	
Voice Controlled Ledger	Mar 2026
- Architected an event-driven 3-model AI pipeline (Whisper, GPT, Claude) on containerized FastAPI at sub-3s speech-to-ledger latency - Cut manual ledger-entry time 80% with an asynchronous NLP interface parsing amount and FX across 99 languages - Integrated 5 Google APIs (OAuth 2.0, Sheets, Docs, Calendar) behind a unified, fault-tolerant token-refresh service with least-privilege scoping	

Experience

Cornell Trading <i>Quantitative Chair</i>	Ithaca, NY <i>Aug 2026 – Present</i>
- Designed and taught an 8-week quant curriculum on Python, statistical arbitrage, and backtesting to 43 new members - Directed 6 research teams and standardized a shared Python backtesting framework, cutting strategy-development time 30% - Mentored members through a paper-trading competition; top teams reached a 1.5 Sharpe over 10 weeks	
Fry's Food Stores <i>Operations Intern – Supply Chain Management</i>	Scottsdale, AZ <i>June 2026 – Aug 2026</i>
- Raised ordering efficiency 18% by deploying an AI agent that autonomously reordered out-of-stock products from POS data - Cut void transactions per shift 40% by training 20+ associates on tender, coupon, and price-check standards - Drove till variance under \$2/shift by auditing 5+ tills daily across 3 registers	
Tiny Smiles <i>Risk Management Intern</i>	Scottsdale, AZ <i>May 2025 – Aug 2025</i>
- Ran vectorized Monte Carlo risk simulations on six-figure investment accounts, quantifying VaR and drawdown distributions - Reduced downside-risk exposure 18% through performance attribution and asset-allocation optimization in Excel/VBA	

Teaching Experience

Cornell University – SYSEN 6170 <i>Teaching Assistant – Quantitative Foundations of Systems Engineering</i>	Ithaca, NY <i>Aug 2026 – Present</i>
- Led office hours and recitations for 30 PhD students, resolving 100+ convex-optimization questions - Graded problem sets and modeling case studies on Python/MATLAB optimization formulations; mentored 8 teams through capstone model formulation, validation, and policy analysis	
Cornell University – CEE 5980 <i>Teaching Assistant – Decision Framing and Analytics</i>	Ithaca, NY <i>Aug 2025 – Present</i>
- Supported 30+ graduate students in office hours and graded homework and quizzes on decision trees and multi-objective analytics - Advised teams on decision-analysis capstones spanning problem framing, alternative generation, and triple-bottom-line evaluation	

Technical Skills

Languages: Python, Java, C/C++, SQL, Bash, MATLAB
Tools & Platforms: Git, Perforce, Linux, Docker, Vercel, Render, Jupyter, Excel/VBA
Web & Backend: FastAPI, React, Vite, REST APIs, OAuth 2.0
Libraries: NumPy, Pandas, scikit-learn, statsmodels, PyTorch, Hugging Face Transformers
Quant/ML: Monte Carlo simulation, Hidden Markov Models, Kalman filtering, statistical arbitrage, cointegration & z-score signals, options pricing (Black-Scholes, GBM), backtesting, Random Forests, time-series modeling, convex & linear optimization; AIME Qualifier